

Solar Panel Protection System

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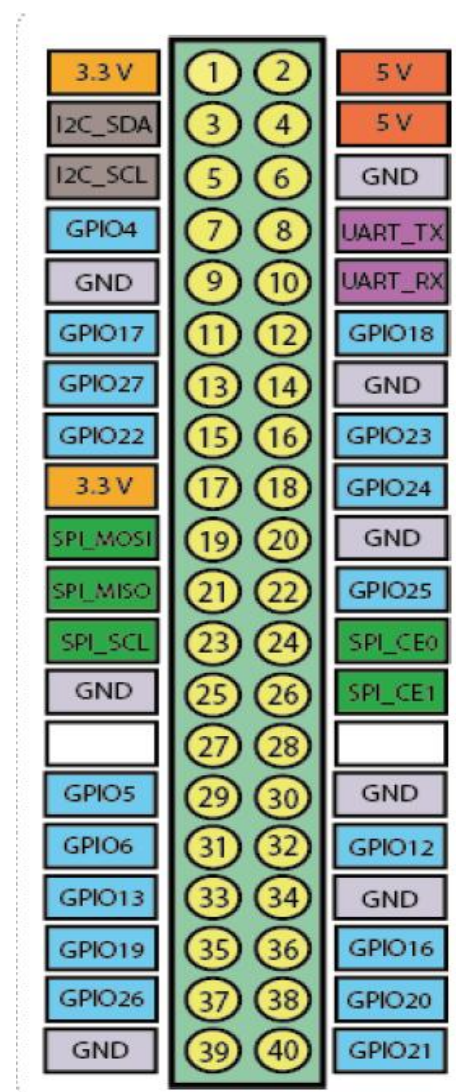
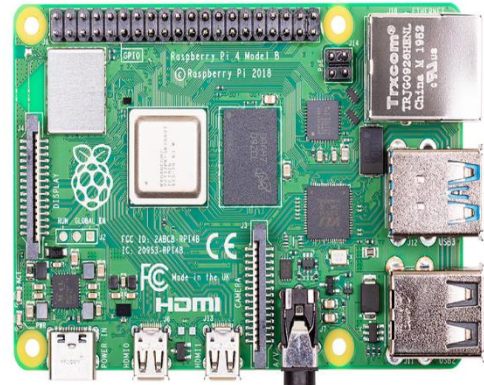
2024

Components:

Raspberry Pi 3 Includes:

- SoC Broadcom Chip containing CPU, RAM, GPU.
- HDMI Output : Used to connect Raspberry Pi to screens, outputs both video and audio.
- Audio Jack (3.5mm) : For connecting headphones or speakers.
- Ethernet Port : For connecting Raspberry Pi to the network.
- USB Ports : For connecting USB devices such as keyboard and mouse.
- Micro USB Power Port : Supplies power to Raspberry Pi at 5 volts.
- SD Card Slot : For inserting the memory card that contains the operating system and files.
- Camera Interface (CSI) : For connecting high-resolution cameras specific to Raspberry Pi.
- Display Interface (DSI) : For connecting touchscreens specific to Raspberry Pi.
- General Purpose Input/Output (GPIO) Pins :
These are the pins on the Raspberry Pi board (40 in total in the latest version). They can be used as inputs or outputs, or for providing power or communication protocols between various electronic devices.

The adjacent figure shows the input/output ports, power ports, and communication protocols such as UART, SPI, and I²C.

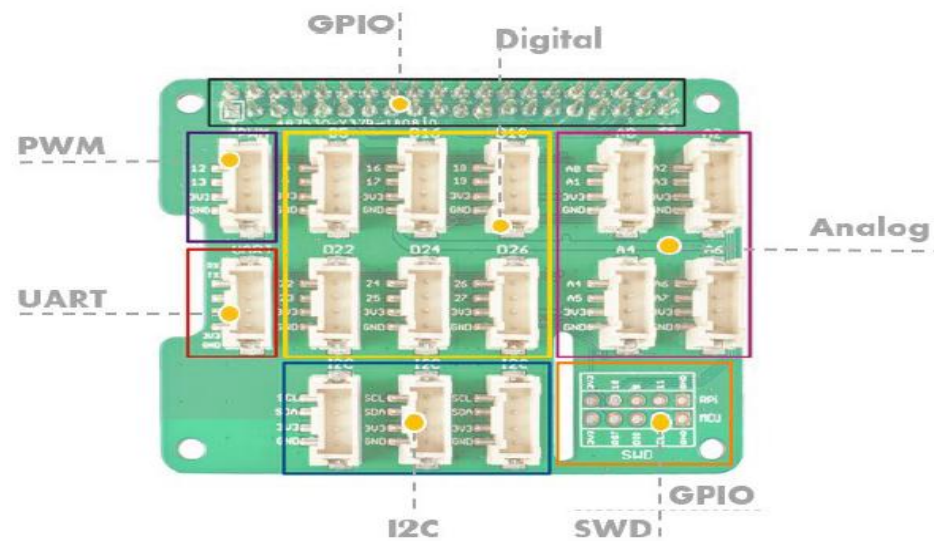


GROVE HAT Board

An expansion board designed to provide an easy interface for connecting various electronic devices and sensors to the Raspberry Pi.

It supports

Digital/Analog/I²C/PWM/UART ports.



Operating voltage: 3.3V

Includes:

GPIO ports

PWM port

SWD port

Digital and Analog ports

UART port: for connecting UART-based sensors and devices

SPI port: for communicating with SPI-based displays

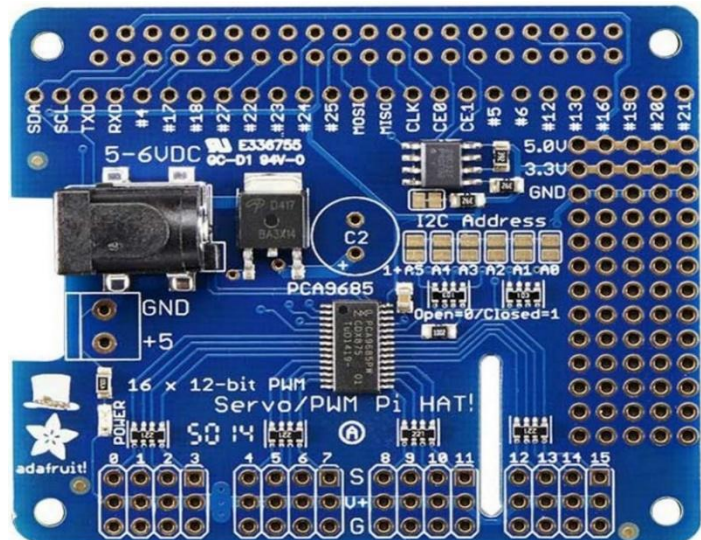
I²C ports (SCL and SDA): used to connect GROVE sensors using I²C protocol to the Raspberry Pi

ADAFRUIT SERVO MOTOR HAT Board

An expansion board used to easily control servo motors.

Contains:

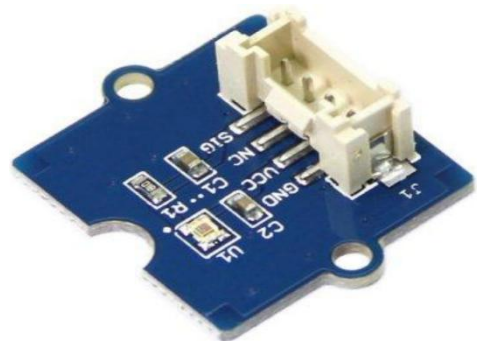
- 16 Servo Ports (SERVO 0 – SERVO 15)
- I²C Interface (SDA, SCL) for communication with the Raspberry Pi
- Additional ports A0 – A5 for connecting extra sensors
- Separate power connector: 5V–6V



Servo Motors

2 motors used to precisely control rotation angle or movement in mechanical systems.

- Powered by an electrical source, typically operates at 4.8V–6V (5.5V used here)



GROVE Luminance Sensor

A sensor used to measure ambient light intensity.

- Operating voltage: 2.4V–5.5V
- Measures light between 0–1000 Lux
- Includes:
 - GROVE interface for connection
 - APDS-9002 chip for measuring generated current

Solar Panels

A group of connected cells that receive sunlight and convert it into voltage, causing electric current to flow.



Motor Shield Board

A board that provides an interface between Raspberry Pi and DC motors.

Supports speed and direction control of motors

Provides overcurrent and short-circuit protection

Based on the L298 chip

Supports two channels (A & B), each capable of handling up to 2 amps

Includes:

DIRA, DIRB: Direction control signals

PWMA, PWMB: PWM signals for speed control

SNSA, SNSB: Current sensing

BRAKE-A, BRAKE-B: Brake control

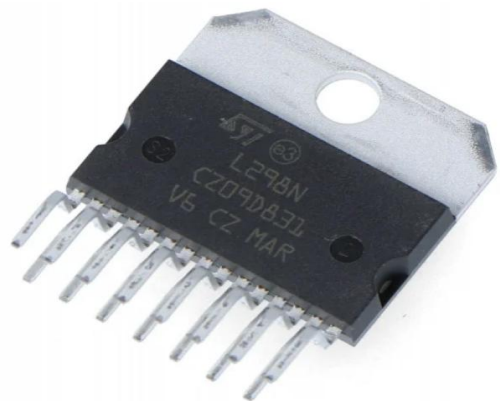
Supports motor voltages from 4.8V–35V

Each channel has LEDs indicating direction of rotation

L298 Chip

A motor driver chip for controlling DC and stepper motors.

- **Can drive multiple motors independently**
- **Overcurrent and short-circuit protection**
- **Operating voltage: 5V**
- **Input/Output ports for motor control**



Grove Infrared Proximity Sensor

An infrared sensor used to detect nearby objects (range: 10cm–80cm).

- Continuously senses its surroundings
- Sends a signal when an object is detected
- Has VCC, GND, and SIGNAL pins
- Compatible with GROVE HAT via GROVE interface



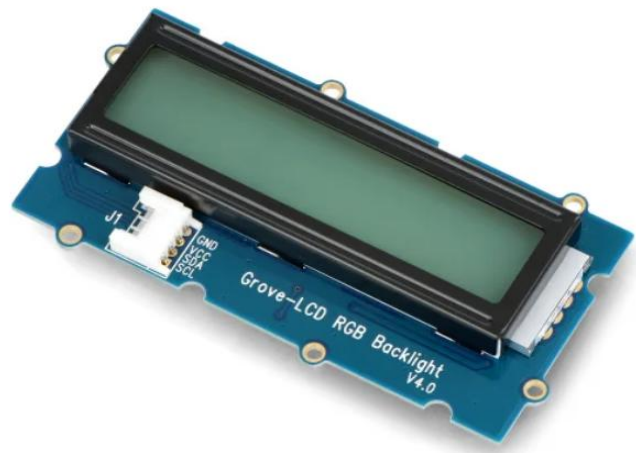
Raspberry Pi Button Breakout Board

A board used to connect push buttons and switches to Raspberry Pi through GPIO pins.

LCD RGB Backlight Display

A high-quality 2x16 character LCD display with RGB backlight.

- Uses the JHD-2X16-I2C chip
- Has 4 pins: VCC, GND, SCL, SDA for I²C communication



Wiring Overview:

- GROVE HAT connects to Raspberry Pi via I²C
- Servo Motor connects to Adafruit Servo Motor Hat via Servo 0, which connects to Raspberry Pi via I²C
- Two Luminance Sensors connect to AD0 and AD1 on GROVE HAT
- Infrared Proximity Sensor connects to AD2 on GROVE HAT
- Button Breakout Board connects to GPIO25
- LCD RGB Display connects to Raspberry Pi via I²C
- Motor Shield connects to Raspberry Pi via several GPIO pins:
 - PWMA (GPIO4), PWMB (GPIO20)
 - DIRA (GPIO5), DIRB (GPIO21)
 - BRAKE-A (GPIO6), BRAKE-B (GPIO22)
 - SNSA (GPIO17), SNSB (GPIO18)

System Operation:

In Case of Theft (Object Approaching Solar Panels):

- The infrared sensor continuously reads voltage.
- If voltage > 2: someone is near the panels → "Caution! Please Check" appears on LCD.
 - If the system is locked, nothing happens.
 - If unlocked, the system locks (panels covered and access blocked).
- If voltage < 2: normal condition → wait for button press.
 - Pressing the button while system is unlocked → lock system → message: "Orders to cover P1&P2"
 - Pressing the button while system is locked → unlock system → message: "Orders to uncover P1&P2"

In Case of High Brightness (High Illumination):

- Luminance sensor continuously measures brightness and displays values (LUX1, LUX2) on LCD.
- When LUX > 900: solar panels are covered to protect them from overheating.
 - Message: "Covering P1 TOO HOT weather" / "Covering P2 TOO HOT weather"
- If LUX > 900 and panels are already covered → no change
- If LUX < 900 and panels are covered → uncover them
- If LUX < 900 and panels are uncovered → continue monitoring

Metal covers are used to protect the solar panels.